

POKHARA UNIVERSITY

Level: Bachelor
Programme: BCA

Semester: Spring

Course: Computer Graphics and Multimedia Technology
(New)

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is the difference between computer graphics and image processing? Explain the Application of computer graphics. 7
b) List out the merits and demerits of penetration techniques? Explain color generation technique in CRT monitor. 8
2. a) Digitize the necessary pixel positions of a line between starting coordinate (20,10) and ending coordinates (26,15) by using Bresenham's line drawing algorithm. 7
b) Define Scan Conversion. Derive midpoint circle algorithm for drawing a circle. 8
3. a) Why do we need clipping? Explain Cohen-Sutherland Line Clipping algorithm. 8

OR

Find the transformation matrix the transforms that rectangle ABCD whose center is at (4,2) is reduced to half of its size, the center will remain same. The co-ordinate of ABCD are A (0,0), B (0,4), C (8,4) and D (8,0). Find Coordinate of new, square. Also derive the transformation matrix to convert this rectangle to square.

- b) Prove the statements: 7
 - i. Two successive rotations are always additive
 - ii. Two successive translations are always additive
4. a) Define projection. Explain perspective projection with the expression and matrix representation. 7

OR

How Geometric tables are used to represent a 3D object? Explain with example. Give conditions to generate error free table.

- b) Why is it necessary to remove hidden surface? Explain Z-Buffer in detail. 8

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| 5. | a) | Define the term Rendering. How Phong shading is better than Gouraud shading to calculate intensities. Explain. | 8 |
| | b) | What is Multimedia Authoring? List different Multimedia Authoring tools and explain any two. | 7 |
| 6. | a) | What is MIDI? Explain the MIDI hardware, software and communication between them. | 8 |
| | b) | How video works? Explain various video compression techniques. | 7 |
| 7. | Write short notes on: (Any two) | | 2×5 |
| | a) | Illumination Models | |
| | b) | Types of Animation | |
| | c) | Image formats | |